

Explainable AI–Based Detection of Pediatric Cardiovascular Diseases

Using Multi-Lead ECG Signals

Problem Statement & Motivation

Pediatric cardiovascular diseases often present subtle and age-dependent electrocardiographic patterns that are difficult to interpret consistently using conventional diagnostic approaches. While recent advances in artificial intelligence have demonstrated promising results in ECG-based disease detection, most existing models are adult-centric and operate as black boxes, offering limited interpretability. There is a critical need for pediatric-specific, explainable AI models.

Electrocardiography is a widely used, non-invasive diagnostic tool in pediatric cardiology; however, ECG interpretation in children is more complex than in adults due to developmental changes in cardiac physiology and waveform morphology. Integrating explainable AI techniques into this analysis is essential to bridge the gap between model predictions and clinical reasoning.

Project Objectives

- To preprocess and analyze multi-lead pediatric ECG signals for selected cardiovascular diseases.
- To develop supervised deep learning models for detecting pediatric cardiovascular diseases from ECG data.
- To apply explainable AI techniques to interpret model predictions at the ECG waveform level.
- To study age-dependent variations in ECG-based disease detection and model explanations.
- To evaluate model performance, robustness, and failure cases in clinically relevant scenarios.

Expected Deliverables

- Cleaned and curated pediatric ECG subsets for selected diseases.
- Trained deep learning models for disease classification.
- Explainability visualizations highlighting ECG regions influencing predictions.
- Age-wise comparative analysis of model behavior and explanations.
- A complete experimental pipeline with documented code and results.

- A manuscript-ready set of figures, tables, and methodological descriptions suitable for Q1 journal submission.

Detailed Weekly Schedule

Week & Focus	Goal	Tasks & Expected Deliverables
Week 0 Orientation	Conceptual grounding (no coding)	Tasks: Study ECG fundamentals (P, QRS, T waves), myocarditis/congenital heart disease effects. Deliverable: 1–2 page typed explanation.
Week 1 Dataset Familiarization	Remove fear of data	Tasks: Download dataset, plot one lead, visually identify waves, learn about leads and labels. Deliverable: Notebook with plotted ECG and disease label explained.
Week 2 Signal Understanding	Biomedical interpretation	Tasks: Plot and visually compare ECG from a healthy child vs. diseased child. Read 1-2 papers. Deliverable: Side-by-side ECG plots and written observation.
Week 3 Feature Awareness	Build intuition (no DL)	Tasks: Extract HR, RR interval, QRS duration. Explain medical significance. Deliverable: Table of extracted features and biomedical explanation.
Week 4 First ML Model	Confidence building	Tasks: Train simple CNN for binary classification. Observe behavior, do not celebrate accuracy. Deliverable: Metrics, confusion matrix, and short reflection.
Week 5 Explainable AI	Core contribution	Tasks: Apply Saliency map or Grad-CAM. Overlay on ECG. Deliverable: Explained ECG plots and visual interpretation.
Week 6 Age-Wise Analysis	Pediatric uniqueness	Tasks: Split data by age, repeat inference, compare explanations. Deliverable: Age-wise comparison plots and observation notes.

Week & Focus	Goal	Tasks & Expected Deliverables
Week 7 Error & Failure Analysis	Reviewer-proofing	Tasks: Identify cases where disease ECG looks normal and model fails. Explain clinically. Deliverable: Failure case discussion, screenshots, and reasoning.
Week 8+ Consolidation	Paper-ready material	Tasks: Compile figures, tables, flowchart, explanation visuals. Deliverable: Finalized manuscript components targeting Q1 journal.

Outcome at Project Completion

At the end of the project, the student will have developed a strong biomedical and computational understanding of pediatric ECG analysis, produced interpretable AI-based disease detection results, and contributed substantially to a research study suitable for submission to a high-quality (Q1) journal. The project emphasizes scientific reasoning, explainability, and clinical relevance rather than mere model accuracy, aligning with current expectations in medical AI research.